Numerical Lab #1

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The question below matches Exercise 5 in the IPython notebook. Solve it and write your code and comments in the space provided in the notebook. You may search the Internet, ask your classmates, or ask the instructor for help. You may not ask the instructor for the answer. Do not copy the answer line by line from the Internet or from your classmates.

The Taylor polynomial of degree n for $f(x) = e^x$ is $\sum_{i=0}^{n} \frac{x^i}{i!}$. Use the Taylor polynomial of degree 9 and **apply three-digit rounding arithmetic to each number in your computations** to find an approximation to e^{-5} using each of the following methods:

(a) $$e^{-5} \approx \sum_{i=0}^{9} \frac{(-5)^i}{i!} = \sum_{i=0}^{9} \frac{(-1)^i 5^i}{i!}.$$

(b) $$e^{-5} = \frac{1}{e^5} \approx \frac{1}{\sum_{i=0}^{9} \left(\frac{5^i}{i!} \right)}.$$

Assume e^{-5} true value is 0.0067379470. Which formula, (a) or (b), gave the most accuracy, and why? **For your chosen formula, propose a way to become a bit better**.